

Cyclic-Group Composition as a Diagnostic for Architectural Inductive Bias:

Why Complex Unit-Circle Networks Generalize Where Real-Valued Gated Recurrences Do Not

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Abstract

We ask when, if ever, complex-valued neural architectures offer inductive-bias advantages over real-valued ones. To make the question crisp we study cyclic-group composition tasks: sequences containing one of n atoms and k instances of a single “rotation” token, with label $(c + k) \bmod n$, training on $k \in \{0, \dots, K_{\text{tr}}\}$ and evaluating on k up to $4K_{\text{tr}}$. Across five experiments and 200+ training runs we find: (i) A 303,000-parameter complex transformer with Hermitian attention learns a phase-flip representation of the rotation token (77% of probe pairs near π on $\mathbb{Z}/2$) but cannot compose it—softmax weights sum to one, forcing additive composition. (ii) A 273-parameter set-equivariant phase-sum network (PHASESUMNET) generalizes perfectly to out-of-distribution depths on $\mathbb{Z}/n$ for every $n \in \{2, 3, 5, 7, 11, 13\}$ tested, with zero variance across seeds. (iii) Real-valued gated recurrences (GRU, LSTM) solve $\mathbb{Z}/2$ and $\mathbb{Z}/3$ but collapse to chance ($1/n$) on $\mathbb{Z}/5$ across a $285\times$ capacity sweep from 4k to 1M parameters. The mechanism is state saturation against the tanh ceiling, after which the linear readout becomes a near-constant function of k . (iv) On $\mathbb{Z}/7$ with two-layer GRU at large width, the barrier *leaks via grokking*: a late-training phase transition (loss already at zero) sometimes finds an approximately periodic state trajectory that partially aligns with the readout, reaching OOD accuracy 0.95. Across every condition, the complex unit-circle architecture solves the task two-to-three orders of magnitude more efficiently in parameters, with no phase transition. We argue that the qualitative separation is between architectures whose composition operator is a group homomorphism by construction and those that must discover one through optimization.

1 Introduction

A familiar story in deep learning attributes empirical success to flexible function approximators trained on enough data with enough compute. A second story, less often told but equally important, is that the architecture’s *built-in symmetries* determine what it can generalize to from finite data: a convolutional network generalizes across spatial translations because translation-equivariance is hardwired; a permutation-invariant set network generalizes across input orderings because it cannot tell them apart in the first place; a transformer with rotary positional encoding generalizes to unseen positions roughly to the extent that its relative-position construction is faithful to a continuous group action.

This paper asks a sharp version of the second question. *Given a synthetic task whose semantics is governed exactly by the cyclic group $\mathbb{Z}/n\mathbb{Z}$, which architectures generalize across the group, and why?* The question has a known partial answer: architectures that *represent* the group action—either by construction or by training—generalize, while architectures that do not, do not. The

contribution of this paper is to make this folk wisdom mathematically precise on a controlled task family, characterize the failure modes of several common architectures (transformer, GRU, LSTM, additive linear), and identify a previously-unnoticed phenomenon: even when an architecture is in principle capable of representing the group, doing so may require an optimization-time phase transition (grokking) that does not occur for natively-symmetric architectures.

1.1 Contributions

1. We formalize a cyclic-group composition task on $\mathbb{Z}/n\mathbb{Z}$ and prove that any additive architecture composing token embeddings with a linear readout produces logits linear in the rotation count k , which cannot represent $(c + k) \bmod n$ for $n \geq 3$. The model provably converges to uniform output (Theorem 3).
2. We exhibit a minimal architecture, PHASESUMNET, that represents the group action by construction via unit-circle phases. It achieves OOD accuracy 1.000 ± 0.000 across $n \in \{2, 3, 5, 7, 11, 13\}$ with 306–845 parameters.
3. We show that softmax-attention transformers learn the single-token group action (probe: 77% of rotation pairs near π on $\mathbb{Z}/2$) but fail to compose it, because softmax weights summing to one make attention an additive composition operator (Proposition 7).
4. Across a capacity sweep of GRU and LSTM from 4,000 to 1,060,000 parameters, real-valued gated recurrence fails on $\mathbb{Z}/5$ at chance ($1/n$), with the mechanism identified by trajectory analysis: tanh-bounded state saturates past training depth, producing constant logits.
5. On $\mathbb{Z}/7$ with two-layer GRU at large width, we observe *grokking*: a phase transition during late training where the model’s OOD accuracy jumps from ~ 0.70 to ~ 0.95 while training loss has been zero for thousands of steps. The grokked model’s state has stopped saturating; its trajectory diagnostics place it precisely between the failing baseline and the unit-circle reference.

1.2 Roadmap

Section 2 reviews related work on complex networks, group-equivariant architectures, length generalization, and grokking. Section 3 develops the theoretical framework: cyclic-group composition tasks, the question of architectural representability, and the negative theorem for additive-linear architectures. Sections 4 and 5 describe the task family and the six architectures we study. Sections 6 to 10 report the five experiments. Section 11 performs trajectory analysis to identify the GRU failure mechanism and the $\mathbb{Z}/7$ grokking phenomenon. Section 12 discusses what the findings collectively imply, with explicit limitations.

2 Background and Related Work

Complex-valued neural networks. Trabelsi et al. (2018) provide a comprehensive blueprint for training complex-valued networks at scale, including modReLU, complex batch normalization, and complex initialization schemes. Earlier, Arjovsky et al. (2016) proposed unitary RNNs whose recurrent transitions are constrained to lie on the unitary group, arguing that the resulting orbit structure stabilizes long-range dependencies. Sordani et al. (2013) applied complex amplitudes to language modeling in an information-retrieval setting. Despite this body of work, complex-valued networks have remained niche; one recurring critique is that they have not produced demonstrable advantages over real-valued networks at matched parameter counts on standard benchmarks.

Group-equivariant architectures. Cohen and Welling (2016) formalized group equivariance for CNNs, extending to non-Euclidean groups in subsequent work. The unifying principle—that an architecture should commute with the symmetry group acting on its input space—has produced equivariant networks for SE(3) molecular dynamics, SO(3) point clouds, and others. The cyclic group $\mathbb{Z}/n$ is the simplest non-trivial group with finite order, and serves here as a minimal benchmark for the broader equivariance question.

Length and depth generalization in sequence models. Hahn (2020) proves that softmax self-attention cannot recognize certain regular languages—specifically parity—in the infinite-length limit, a result we use to interpret our $\mathbb{Z}/2$ transformer experiment. Bhattamishra et al. (2020) empirically confirms that transformers fail to learn parity at depths beyond their training distribution. Our $\mathbb{Z}/n$ task family generalizes parity to arbitrary cyclic groups and provides a graded difficulty spectrum.

Grokking. Power et al. (2022) introduced the term *grokking* for a training pattern in which a network reaches near-perfect training accuracy quickly but generalization on a held-out set emerges only after a long period of further training. Subsequent work (Nanda et al., 2023; Liu et al., 2022) has connected grokking to the slow discovery of low-norm or structured solutions in overparameterized models. Our $\mathbb{Z}/7$ result connects grokking to the specific question of whether an architecture’s gradient descent can *discover* a group-respecting representation late in training, absent that representation being built in.

Length generalization on algorithmic tasks. Anil et al. (2022); Zhou et al. (2023); Jelassi et al. (2024) report that transformers struggle with length generalization on algorithmic tasks (parity, addition, copying) and analyze the specific failure modes. Our work is in the same lineage but uses a cleaner $\mathbb{Z}/n$ family to isolate the algebraic structure from positional encoding details.

3 Theoretical Framework

3.1 Cyclic-Group Composition Tasks

Fix $n \in \mathbb{N}_{\geq 2}$. The *cyclic-group composition task* on $\mathbb{Z}/n\mathbb{Z}$ is defined by the following data-generating process. The input alphabet Σ consists of:

- n *atom* tokens $a_0, a_1, \dots, a_{n-1}$, each carrying a class label;
- one *rotation* token τ ("TWIRL");
- F semantically inert *filler* tokens $f_1, \dots, f_F$.

A sample $\mathbf{x} = (x_1, x_2, \dots, x_L)$ is constructed by choosing an atom class $c \in \{0, \dots, n-1\}$ uniformly at random, a rotation count k from a specified distribution, and a filler count ϕ , then placing a_c together with k copies of τ and ϕ fillers in a uniformly random order. The label is

$$y(\mathbf{x}) = (c + k) \bmod n.$$

The task is order-invariant by construction: y depends only on which atom is present and how many rotations there are.

Concretely, training samples draw k uniformly from $\{0, 1, \dots, K_{\text{tr}}\}$; evaluation samples are *depth-stratified*: equal numbers at each $k \in \{0, 1, \dots, K_{\text{eval}}\}$ with $K_{\text{eval}} \gg K_{\text{tr}}$. A model that has learned the group structure should achieve $\sim 100\%$ accuracy on every k . A model that has

learned a lookup table over training depths will achieve 100% on $k \leq K_{\text{tr}}$ and degrade on the out-of-distribution (OOD) depths $K_{\text{tr}} < k \leq K_{\text{eval}}$.

3.2 What Does It Mean for an Architecture to “Represent” the Group?

Definition 1 (Group-respecting architecture). Let $\mathcal{A}$ be a sequence-to-class architecture mapping inputs $\mathbf{x} \in \Sigma^*$ to logits $\ell(\mathbf{x}) \in \mathbb{R}^n$. We say $\mathcal{A}$ is *group-respecting* for $\mathbb{Z}/n$ on the composition task if there exists a representation $\rho : \mathbb{Z}/n \rightarrow \text{GL}(V)$ on some real or complex vector space V , a class embedding $\phi : \{a_0, \dots, a_{n-1}\} \rightarrow V$, a filler-invariant state space $\mathcal{H}$, and a readout $R : V \rightarrow \mathbb{R}^n$ such that the architecture’s computation is equivalent to

$$\ell(\mathbf{x}) = R(\rho(k) \phi(c)), \quad (1)$$

where c, k are the atom class and rotation count of $\mathbf{x}$, and the readout satisfies the homomorphism property $R(\rho(k + \Delta)v) = R(\rho(\Delta) \rho(k)v)$ for all k, Δ, v .

The architectural separation we study is whether Equation (1) holds *by construction* (some architectures), can be *discovered* by training (others), or is provably unattainable regardless of parameters and training (Theorem 3 below).

3.3 The Negative Theorem for Additive-Linear Architectures

We now state the result that determines the lower bound of the spectrum.

Definition 2 (Additive-linear architecture). An architecture is *additive-linear* if its logits take the form

$$\ell(\mathbf{x}) = W \left(\sum_{t=1}^L e(x_t) \right) + b \in \mathbb{R}^n,$$

for some embedding $e : \Sigma \rightarrow \mathbb{R}^d$, readout matrix $W \in \mathbb{R}^{n \times d}$, and bias $b \in \mathbb{R}^n$.

Theorem 3 (Additive-linear cannot represent $\mathbb{Z}/n$ above chance). *Let $\mathcal{A}$ be an additive-linear architecture and let $y(\mathbf{x}) = (c + k) \bmod n$ be the cyclic-group composition task on $\mathbb{Z}/n\mathbb{Z}$ with $n \geq 2$. There is no choice of e, W, b such that $\arg \max_j \ell_j(\mathbf{x}) = y(\mathbf{x})$ for all sufficiently long k trajectories. In particular, no additive-linear model can achieve OOD accuracy above $1/n$ on the depth-stratified evaluation distribution with $K_{\text{eval}} > 2n$.*

Proof. Fix the atom class c and let the filler contribution be folded into a constant offset (we may assume the optimal model assigns $e(f_i) = 0$ for all fillers, since fillers carry no signal). For an input with exactly k rotation tokens the logits become

$$\ell_j(c, k) = W_{j,:} (e(a_c) + k \cdot e(\tau)) + b_j = \alpha_j(c) + \beta_j \cdot k,$$

with $\alpha_j(c) = W_{j,:} e(a_c) + b_j$ and $\beta_j = W_{j,:} e(\tau)$, both independent of k . Each class logit is therefore an affine function of k with slope β_j that does not depend on the atom class c .

For the model to predict class j at rotation count k we need $\ell_j(c, k) > \ell_{j'}(c, k)$ for all $j' \neq j$. This is the condition $(\beta_j - \beta_{j'})k > \alpha_{j'}(c) - \alpha_j(c)$ for all $j' \neq j$, a system of $n - 1$ linear inequalities in k . If $\beta_j > \beta_{j'}$ this holds for all sufficiently large k ; if $\beta_j < \beta_{j'}$ it holds for all sufficiently small k . Therefore there is at most one class j^* (the one with largest slope β_{j^*}) for which ℓ_{j^*} becomes the argmax for all large enough k .

The correct label, however, cycles: $y(c, k) = (c + k) \bmod n$ takes every value in $\{0, 1, \dots, n - 1\}$ as k ranges over any window of length n . For $K_{\text{eval}} > 2n$, the correct label takes each of n values on (approximately) $1/n$ of the OOD depths. The model predicts j^* on all of them. Hence the model is correct at most $\frac{1}{n}$ of the time on sufficiently long OOD trajectories. $\square$

Remark 4 (Topology). The mechanism behind Theorem 3 is the topology of $\mathbb{R}$: at most $n - 1$ pairwise-line crossings can occur between any n affine functions of a single variable, so the argmax has at most $n - 1$ transitions and is eventually constant in k . The unit circle in $\mathbb{C}$ avoids this restriction precisely because it has no point at infinity, and a continuous trajectory along it can visit all n “slots” an unbounded number of times.

Remark 5 (Strengthening). The proof in fact rules out *any* architecture whose logits are an affine function of token count k in some specified token, regardless of how nonlinear the upstream embedding is. The bound extends to in-distribution accuracy whenever $K_{\text{tr}} \geq n - 1$: the same argmax-transition bound applies to the union of ID and OOD points. Inspecting the trained REALADDNET on $\mathbb{Z}/3$ ($K_{\text{tr}} = 5 > 2$), the per-class slope spread $\max_j \beta_j - \min_j \beta_j$ is 0.020 ± 0.005 across seeds, indicating the optimizer converged to nearly-equal slopes and effectively uniform logits—consistent with the theorem-bounded failure.

Remark 6 ($n = 2$ caveat). At $n = 2$, the bound $1/n = 1/2$ numerically coincides with chance, so the failure is less visually dramatic than for $n = 3$ (chance 0.333) or $n = 13$ (chance 0.077). We measure REALADDNET OOD accuracy of 0.505 ± 0.015 on $\mathbb{Z}/2$, which is at chance and consistent with the theorem. The representational impossibility is the same; only the visibility of the failure differs.

3.4 Softmax Attention is Additive Composition

The same argument extends to a broad class of attention-based architectures.

Proposition 7 (Softmax attention as a stochastic convex combination). *Let an attention layer compute its output at query position q as $z_q = \sum_t \alpha_{q,t} v(x_t)$, where the attention weights $\alpha_{q,t} \geq 0$ satisfy $\sum_t \alpha_{q,t} = 1$, and v is the value projection. Then z_q is a convex combination of value vectors, and its magnitude is bounded by $\max_t \|v(x_t)\|$.*

The proposition is a consequence of the softmax constraint and is not novel. Its relevance here is the following corollary, which makes precise what Hahn (2020) showed for parity in the formal-language setting.

Corollary 8 (Attention cannot multiply). *Let $\rho : \mathbb{Z}/n \rightarrow \text{GL}(V)$ be a faithful representation acting by multiplication: $\rho(k_1 + k_2) = \rho(k_1)\rho(k_2)$. Suppose an architecture $\mathcal{A}$ computes, at its output position, a single softmax-attention aggregation over per-token value vectors $v(x_t)$ followed by a linear readout. Then $\mathcal{A}$ cannot represent $\rho(k)v$ for arbitrary k via the attention output, because the attention output is a convex combination of the $v(x_t)$ values, which lies in their convex hull, while $\rho(k)v(x_t)$ ranges over orbits in V that exit any convex hull for sufficiently large k in general.*

In short: attention can mix tokens, but it cannot multiply them. The complex-valued attention we test in Section 6 learns the correct per-token rotation $\rho(\tau) \approx -I$ in its embedding, but the architecture’s composition operator (attention) is unable to use it.

3.5 What Unit-Circle Phases Buy

The simplest architecture satisfying the homomorphism condition Equation (1) for arbitrary $\mathbb{Z}/n$ uses the unit circle in $\mathbb{C}$ as its state space.

Definition 9 (Phase-sum architecture). Let each token $x \in \Sigma$ be assigned a phase vector $\theta(x) \in \mathbb{R}^d$. The phase-sum architecture computes

$$\Theta(\mathbf{x}) = \sum_{t=1}^L \theta(x_t) \in \mathbb{R}^d, \quad \ell(\mathbf{x}) = R(\cos \Theta(\mathbf{x}), \sin \Theta(\mathbf{x})), \quad (2)$$

where R is a linear readout $\mathbb{R}^{2d} \rightarrow \mathbb{R}^n$, and $\cos, \sin$ are applied componentwise.

Proposition 10 (Phase-sum is group-respecting for every $\mathbb{Z}/n$). *For any $n \geq 2$, the phase-sum architecture admits a parameter setting with $d = 1$ such that $\ell(\mathbf{x}) = R'(\rho(k) \cdot \phi(c))$ in the sense of Definition 1, where $\rho(k)$ acts on $\mathbb{C}$ by multiplication by $e^{2\pi i k/n}$.*

Proof. Choose $\theta(a_c) = 2\pi c/n$ for $c = 0, \dots, n-1$, $\theta(\tau) = 2\pi/n$, and $\theta(f_i) = 0$ for all fillers. Then $\Theta(\mathbf{x}) = 2\pi(c+k)/n$, and $(\cos \Theta, \sin \Theta) \in \mathbb{R}^2$ identifies with the point $e^{2\pi i(c+k)/n}$ on the unit circle. The unit circle contains exactly n points $e^{2\pi i j/n}$ for $j = 0, \dots, n-1$, and these are linearly separable in $\mathbb{R}^2$ (any n points on a circle in general position are). A linear readout R' that picks out the correct class from the position on the unit circle exists. $\square$

Remark 11. The proof exhibits a $d = 1$ solution. In practice we initialize with $d = 16$ phase dimensions, providing redundant routes to the optimal solution; the trained network typically realizes the group action only in a subset of dimensions and the readout learns to ignore the rest. We discuss the empirical convergence properties in Section 7.

3.6 The Saturation Failure of Bounded Real-Valued Recurrence

Real-valued gated recurrences (GRU, LSTM) constrain their hidden state to lie in $[-1, 1]^d$ via the tanh activation. This boundedness is usually a virtue (gradient stability), but in our setting it produces a specific failure mode: *saturation* past the training depth.

We will show empirically (Section 11) that for the GRU on $\mathbb{Z}/5$, the hidden-state magnitude $\|h(k)\|$ grows monotonically with rotation count k from ~ 5.12 at $k = 0$ to ~ 7.04 at $k = 30$, asymptoting against the bound. Once saturated, the linear readout's output becomes a near-constant function of k , and the argmax is constant. A constant argmax over n balanced classes scores $1/n$ in expectation. This precisely matches our measured asymptote of ~ 0.25 for $\mathbb{Z}/5$ and ~ 0.14 for $\mathbb{Z}/7$.

We give an informal argument for why this happens. The GRU update $h_{t+1} = (1 - z_t)h_t + z_t\tilde{h}_t$ with $\tilde{h}_t \in (-1, 1)^d$ produces an exponentially-weighted moving average of bounded targets. When the input is a long homogeneous sequence of one token, the candidate $\tilde{h}_t$ becomes approximately constant, and h_t exponentially converges to that fixed point. Across the training distribution (depths $k \leq K_{\text{tr}}$), the model can configure this fixed point to encode useful information about the partial state. Beyond K_{tr} , the convergence to fixed point dominates, and information about $k \bmod n$ that was carried in transient state dynamics is lost.

This argument predicts that an unbounded real-valued recurrence (e.g., a linear RNN with appropriate normalization) might escape the saturation. We do not test this directly in the present paper, but identify it as a target follow-up.

4 The Task Family

We instantiate the cyclic-group composition task with the following parameters across experiments. The vocabulary contains a PAD token, a CLS token (used by the transformer architectures of Section 6), the n atom tokens, the rotation token τ , and 10 filler tokens. Sequences are zero-padded to the maximum length within each batch. The atom appears exactly once per sequence; rotation count k is drawn uniformly from $\{0, \dots, K_{\text{tr}}\}$ during training and from each integer in $\{0, \dots, K_{\text{eval}}\}$ separately during evaluation (depth-stratified). Filler count is drawn uniformly from $\{0, \dots, 8\}$.

The atom and rotations are placed in uniformly random order within the body of the sequence, so positional templates cannot help. This order-invariance is what makes a set-equivariant architecture (Definition 9) the architecturally-correct choice.

The cardinal experiment parameter choices are summarized in Table 1.

Table 1: Task and training parameters across experiments.

Experiment	Group $\mathbb{Z}/n$	K_{tr}	K_{eval}	Seeds
1: complex transformer	$\mathbb{Z}/2$	3	10	5
2: recurrent architectures on $\mathbb{Z}/2$	$\mathbb{Z}/2$	3	15	5
3: $\mathbb{Z}/3$ separation test	$\mathbb{Z}/3$	5	20	5
4: scaling sweep	$\mathbb{Z}/n, n \in \{2, 3, 5, 7, 11, 13\}$	5	20	3
5: capacity sweep	$\mathbb{Z}/5, \mathbb{Z}/7$	5	20	3

5 Architectures

We compare six architectural families, each described below with its relation to the theoretical framework of Section 3.

5.1 Complex Transformer (ComplexTransformer)

A pre-norm transformer encoder operating on $\mathbb{C}^d$ hidden states. Attention uses the Hermitian inner product $\langle q, k \rangle_H = \text{Re}(q^* k)$. Nonlinearity is modReLU (Trabelsi et al., 2018): $\text{modReLU}(z) = \text{ReLU}(|z| + b) \cdot z / (|z| + \varepsilon)$. Positional encoding is $e^{i\omega_d t}$ on each complex dimension. A CLS token’s complex hidden state is read out via $\text{Re}(w^* h_{\text{CLS}})$ for logits. Parameter count is matched to a real-valued baseline transformer of the same depth and head count: 303k parameters at $d = 64$ vs. 310k for the real model at $d = 92$.

By Corollary 8, the architecture cannot multiplicatively compose per-token group actions, regardless of how faithfully the individual token embeddings learn the action.

5.2 Phase-Sum Network (PhaseSumNet)

Definition 9. For task $\mathbb{Z}/n$, the readout is linear $\mathbb{R}^{2d} \rightarrow \mathbb{R}^n$. We use $d = 16$ throughout (giving 16 redundant phase dimensions). Initialization is critical: phases are drawn uniformly from $[-\pi, \pi]$. We discuss why in Section 5.6.

5.3 Real Additive Network (RealAddNet)

The negative-control architecture of Theorem 3:

$$\ell(\mathbf{x}) = W \left(\sum_t e(x_t) \right) + b,$$

with $e : \Sigma \rightarrow \mathbb{R}^{2d}$, $W \in \mathbb{R}^{n \times 2d}$. By the theorem, this architecture cannot exceed $1/n$ accuracy on $\mathbb{Z}/n$ for $n \geq 3$.

5.4 Gated Complex RNN (GatedComplexRNN)

A bidirectional recurrent network with $\mathbb{C}^d$ hidden states. Each token x induces an update

$$\hat{h}_t = \exp(i\theta(x)) h_t + v(x), \quad (3)$$

$$h_{t+1} = (1 - g(x))h_t + g(x)\hat{h}_t, \quad (4)$$

where $\theta(x) \in \mathbb{R}^d$ is a learned per-dimension phase rotation, $v(x) \in \mathbb{C}^d$ a learned complex additive contribution, and $g(x) \in (0, 1)^d$ a sigmoid gate. Composition is multiplicative via the rotation $e^{i\theta(\tau)}$ for the rotation token; the architecture is in principle group-respecting if the model learns $\theta(\tau) = 2\pi/n$ and $g(\tau) \rightarrow 1$, $v(\tau) \rightarrow 0$.

5.5 GRU and LSTM Baselines (GRU, LSTM)

Standard bidirectional single- and two-layer GRU and LSTM, with a final linear readout from concatenated forward/backward final states. Real-valued, tanh-bounded hidden state. These are parameter-matched (within a few percent) to GATEDCOMPLEXRNN in Section 9, then scaled up to $\sim 1\text{M}$ parameters in Section 10.

5.6 Initialization Considerations

For both PHASESUMNET and GATEDCOMPLEXRNN, the phase embedding θ must be initialized uniformly on $[-\pi, \pi]$, not near zero. With near-zero initialization, every sequence produces $\cos \Theta \approx 1$, $\sin \Theta \approx 0$, so logits are nearly constant across inputs and gradient signal is too weak to escape. With uniform initialization, sequences are spread across the unit circle from the start and learning proceeds. We mention this because the convergence depends sensitively on it and because the failure mode (nominally trainable architecture sits at chance forever) is otherwise puzzling.

6 Experiment 1: Complex Transformer on $\mathbb{Z}/2$

Setup. Train both COMPLEXTRANSFORMER ($d = 64$, 303k params) and a parameter-matched real transformer ($d = 92$, 310k params) on $\mathbb{Z}/2$ parity-of-negation: a single atom $\in \{T, F\}$, k NOT tokens, fillers in random order, with label $\text{atom_sign} \cdot (-1)^k$. Train depths $k \in \{0, 1, 2, 3\}$, evaluate $k \in \{0, \dots, 10\}$, 5 seeds, AdamW with cosine-decayed learning rate.

Probe. For the complex model we additionally measure the *phase shift induced by a single NOT*. For paired sequences (s, s') differing only by one extra NOT, we compute the angle $\Delta = \arg(h_{\text{CLS}}(s')/h_{\text{CLS}}(s))$ in each complex dimension. A model that has learned NOT as multiplication by -1 should produce $\Delta \approx \pi$ in every dimension.

Table 2: Experiment 1: 5-seed averages on $\mathbb{Z}/2$. ID = in-distribution accuracy ($k \leq 3$), OOD = out-of-distribution accuracy ($4 \leq k \leq 10$). Probe column applies only to the complex model.

Model	Params	ID acc	OOD acc	Probe: $ \Delta $
Real Transformer	309,765	1.000 ± 0.000	0.488 ± 0.005	—
Complex Transformer	302,978	1.000 ± 0.000	0.445 ± 0.005	2.43 rad (target π)

Results.

Interpretation. Both transformers achieve perfect ID accuracy and fall to near-chance OOD. The complex model is slightly *worse* than the real one. However, the phase probe reveals that the complex model *did* learn the algebra: the mean rotation induced by a single NOT is 2.43 radians (vs target $\pi = 3.14$), and 77% of dimension-paired sequences land within $\pi/8$ of π . A randomly-initialized network gives $|\Delta| \approx 0.17$.

The failure is therefore not in the per-token representation but in composition. By Proposition 7 and corollary 8, softmax attention is forced to mix token contributions additively; multiplying five rotations of π is not the same operation as averaging them with weights summing to one. The complex model has the right algebra embedded in its weights but cannot use it through the attention mechanism.

7 Experiment 2: Recurrent Architectures on $\mathbb{Z}/2$

Setup. Train three architectures whose composition is multiplicative by construction: PHASESUMNET ($d = 16$, ~ 273 params), GATEDCOMPLEXRNN ($d = 24$, $\sim 3,000$ params), and a parameter-matched GRU baseline ($\sim 3,000$ params). Train depths $k \in \{0, \dots, 3\}$, eval to $k = 15$. 5 seeds.

Table 3: Experiment 2: 5-seed averages on $\mathbb{Z}/2$.

Model	Params	ID acc	OOD acc
PHASESUMNET	273	1.000 ± 0.000	1.000 ± 0.000
GATEDCOMPLEXRNN	4,033	1.000 ± 0.000	0.999 ± 0.000
GRU	3,962	1.000 ± 0.000	0.999 ± 0.000

Results.

Interpretation. All three multiplicative architectures generalize essentially perfectly to OOD depth 15— $5\times$ the training distribution—using fewer than 4,000 parameters, while the 303,000-parameter complex transformer of Experiment 1 sat at chance. The structural prior is doing two-to-three orders of magnitude more work than parameter count.

A subtle point: PHASESUMNET’s phase probe is messy. Across seeds, the mean $\cos \theta(\text{NOT})$ is -0.11 (target -1), and only 24% of phase dimensions land within $\pi/8$ of $\pm\pi$. The model does not converge to the textbook optimum. But it does not need to: with $d = 16$ phase dimensions feeding a linear readout, the readout learns to combine the few well-placed dimensions and ignore the rest. PHASESUMNET’s OOD accuracy is exactly 1.000 nonetheless, with zero variance across seeds.

This experiment establishes that the inductive bias from *multiplicative composition* (not specifically complex numbers) solves $\mathbb{Z}/2$. The real-valued GRU achieves the same OOD accuracy as the complex models. To separate "multiplicative composition" from "complex numbers" we need a group whose representation in $\mathbb{R}^*$ is not native. That is the next experiment.

8 Experiment 3: $\mathbb{Z}/3$ as a Separating Task

Why $\mathbb{Z}/3$. The multiplicative group $\mathbb{R}^*$ contains $\mathbb{Z}/2 = \{+1, -1\}$ as a subgroup but *not* $\mathbb{Z}/3$. There is no order-3 element in $\mathbb{R}^*$ under multiplication. To represent $\mathbb{Z}/3$ in a network, one must either (a) use 2D rotations (equivalently, $\mathbb{C}$ on the unit circle), (b) use multi-state gating that simulates a 3-state automaton, or (c) use a non-additive composition operator. Pure additive-linear architectures are provably barred by Theorem 3.

Setup. Architectures: PHASESUMNET3 (the $n = 3$ version of PHASESUMNET), REALADDNET (the negative control of Theorem 3), GATEDCOMPLEXRNN3, and GRU3. Train depths 0–5, eval 0–20, 5 seeds.

Table 4: Experiment 3: 5-seed averages on $\mathbb{Z}/3$. Chance is $1/3$.

Model	Params	ID acc	OOD acc (6–20)	Closure under $3 \cdot \tau$
PHASESUMNET3	355	1.000 ± 0.000	1.000 ± 0.000	1.000
REALADDNET	611	0.336 ± 0.020	0.341 ± 0.008	—
GATEDCOMPLEXRNN3	4,547	1.000 ± 0.000	0.987 ± 0.006	1.000
GRU3	4,503	1.000 ± 0.000	0.999 ± 0.001	0.953

Results.

Interpretation. Theorem 3 is empirically tight. REALADDNET sits at exactly chance ($1/3$) on both ID *and* OOD. The architecture is incapable of fitting even the training distribution, in confirmation of the proof: linear-in- k logits cannot represent $(c + k) \bmod 3$ at any depth. We verify by inspecting the learned readout: the per-class slopes $\beta_j = W_{j,:}e(\tau)$ have spread 0.020 ± 0.005 across seeds, near zero—the model has correctly identified that no consistent linear function of k predicts the label and converged to uniform output.

Multiplicative architectures all generalize. PHASESUMNET3 at 355 parameters reaches OOD 1.000 with zero variance. GATEDCOMPLEXRNN3 and GRU3 reach 0.987 and 0.999 respectively. The GRU’s slight edge at extreme OOD depths (18–20) is interesting and we return to it in Section 9.

This experiment establishes that the relevant axis is *multiplicative vs. additive composition*, with the negative side of the axis provably barred and the positive side generically successful at $\mathbb{Z}/3$. We have not yet separated complex from real within the multiplicative class. That is Experiment 4.

9 Experiment 4: Scaling Sweep over Group Order n

Setup. Sweep $n \in \{2, 3, 5, 7, 11, 13\}$. For each n , train all four multiplicative-class architectures plus the REALADDNET negative control. Train depths 0–5, eval 0–20, 3 seeds per cell. Total: 72 training runs.

Results. Table 5 summarizes by giving OOD accuracy as a function of n . The full per-depth breakdown is in the project repository.

Table 5: Experiment 4: OOD accuracy (mean of 3 seeds) on $\mathbb{Z}/n$ for each n . Bold = near-perfect; underline = collapse to chance.

Model	$n = 2$	$n = 3$	$n = 5$	$n = 7$	$n = 11$	$n = 13$
PHASESUMNET	1.000	1.000	1.000	1.000	1.000	1.000
REALADDNET	0.505	<u>0.341</u>	<u>0.200</u>	<u>0.136</u>	<u>0.069</u>	<u>0.070</u>
GATEDCOMPLEXRNN	0.967	0.979	0.992	0.980	0.816	0.864
GRU	1.000	0.991	<u>0.248</u>	<u>0.241</u>	<u>0.285</u>	<u>0.237</u>
Chance $1/n$	0.500	0.333	0.200	0.143	0.091	0.077

Interpretation. PhaseSumNet is exact across n . OOD accuracy 1.000 ± 0.000 at every n tested, with 306–845 parameters. The architectural symmetry match holds across an order of magnitude in group order. This is the strongest single result of the paper.

RealAddNet matches the theorem’s chance prediction. For $n \geq 3$, OOD accuracy is statistically indistinguishable from $1/n$. The exception at $n = 2$ (OOD $0.505 \approx$ chance for $\mathbb{Z}/2$) is informative: at $n = 2$ the theorem does not apply (two linear functions can change ranking, $n - 1 = 1$ time), but the model still converged to chance. This is a learning failure—an architecture not theoretically barred can still fail to find a workable solution.

GatedComplexRNN declines gracefully with n . OOD ranges from 0.967 at $n = 2$ to 0.816 at $n = 11$, with a clear break-point around $n \approx 8$. The mechanism is that the architecture’s angular margin between adjacent classes is $2\pi/n$, which shrinks as n grows, while the additive value contribution $v(\tau)$ lets state magnitude drift slightly with sequence length; the ratio of drift to margin determines the break.

GRU shows a phase transition, not a scaling decline. GRU goes from 1.000 at $n = 2$ and 0.991 at $n = 3$ to ~ 0.25 at $n \geq 5$ across every tested n , with ID accuracy still ~ 1.000 throughout. The architecture fits the training set perfectly at every n but generalizes only for $n \in \{2, 3\}$. This is not a gradual loss of capacity; it is a binary representability transition we need to investigate further.

We address the GRU’s $n \geq 5$ collapse via a capacity sweep in Experiment 5 and a trajectory analysis in Section 11.

10 Experiment 5: Capacity Sweep on the GRU Collapse

Question. Is the GRU’s $\mathbb{Z}/n_{\geq 5}$ collapse a *structural barrier* (real-valued gating cannot represent the group at any capacity) or merely *sample-inefficient* (more parameters or training would close the gap)?

Setup. Hold $n = 5$ and $n = 7$ fixed. Sweep $(d_{\text{model}}, n_{\text{layers}}) \in \{(16, 1), (32, 1), (32, 2), (64, 1), (64, 2), (128, 1), (128, 2)\}$ giving an 8-cell capacity ladder from $\sim 4\text{k}$ to $\sim 1.06\text{M}$ parameters. Include LSTM as a control with the same gating-family ancestry but different gate equations. Compare to PHASESUMNET at matched d values. Train 120,000 samples for 25 epochs at $\text{lr} = 3 \times 10^{-3}$, batch size 256, 3 seeds per cell. Total: 126 runs.

Results for $\mathbb{Z}/5$. OOD accuracy summary (Table 6):

Table 6: Experiment 5, $\mathbb{Z}/5$: OOD accuracy across capacity. All GRU/LSTM cells sit at or below 0.34; PHASESUMNET is exactly 1.000 at every cell.

Model	Params	$d = 16, L = 1$	$d = 64, L = 1$	$d = 128, L = 2$	$d = 256, L = 1$
GRU	3.7k–797k	0.285	0.247	0.266	0.240
LSTM	4.8k–1.06M	0.288	0.261	0.290	0.251
PHASESUMNET	453–7.2k	1.000	1.000	1.000	1.000

Across 16 GRU+LSTM cells from 3,717 to 1,059,845 parameters ($285 \times$ capacity range), OOD on $\mathbb{Z}/5$ stays in the band $[0.22, 0.34]$. ID is 1.000 ± 0.000 in every cell. The GRU/LSTM architectures fit the training set perfectly at every scale and produce near-noise OOD predictions.

Results for $\mathbb{Z}/7$. 1-layer GRU and all LSTM stay at chance. But 2-layer GRU climbs with capacity:

Configuration	Params	OOD acc
GRU $d = 32, L = 2$	32k	0.402 ± 0.041
GRU $d = 64, L = 2$	127k	0.564 ± 0.079
GRU $d = 128, L = 2$	499k	0.787 ± 0.094
PHASESUMNET	551–8,711	1.000 ± 0.000

The 2-layer GRU at $d = 128$ reaches 78.7% on $\mathbb{Z}/7$ with 499k parameters, partially escaping the barrier. Single-layer GRU and all LSTM at the same scale stay at chance. PhaseSumNet at 551 parameters hits 1.000.

Interpretation. The $\mathbb{Z}/5$ barrier is sharp. $285 \times$ capacity range, two architecture families, two depths each, three seeds each, total 48 runs, none above 0.34 OOD. This establishes as well as a synthetic-task experiment can establish that real-valued gated recurrence does not solve $\mathbb{Z}/5$ at any capacity tested within 25 epochs.

LSTM ties the GRU result. The structural failure is not GRU-specific. Whatever mechanism prevents $\mathbb{Z}/5$ representation acts in both LSTM and GRU. Possibilities: (a) both architectures share the tanh-bounded state space; (b) both architectures share sigmoid gates; (c) both architectures share a per-step linear contribution from input embeddings. Section 11 provides evidence that the bounded state is the key.

The 2-layer $\mathbb{Z}/7$ result is unresolved. The climb $0.40 \rightarrow 0.56 \rightarrow 0.79$ with capacity at 2-layer is reproducible across the three seeds (stderr 0.04–0.09). It is specifically 2-layer GRU, specifically $\mathbb{Z}/7$. We hypothesize and partially confirm in Section 11 that the second layer permits a late-training phase transition—grokking—during which the architecture discovers an approximately group-respecting state trajectory.

11 Mechanism Analysis: State Trajectories

The capacity sweep established the existence of the $\mathbb{Z}/5$ barrier and the $\mathbb{Z}/7$ partial escape. This section identifies what is happening inside the architectures by direct measurement of state and logit dynamics.

11.1 Methodology

For a trained model, we build paired probe sequences differing only in the number of trailing rotation tokens, with the base prefix (CLS, atom, fillers) held identical across k . Across $k = 0, 1, \dots, 30$, we measure:

1. State magnitude $\|h(k)\|$.
2. State modular return: $\|h(k+n) - h(k)\|$ (per fixed base sequence).
3. Logit modular return: $\|\ell(k+n) - \ell(k)\|$.
4. **Argmax invariance**: the fraction of probe sequences satisfying $\arg \max \ell(k+n) = \arg \max \ell(k)$.

Argmax invariance directly measures the property required for OOD generalization: predictions should be invariant under adding n rotation tokens, since $(c+k+n) \bmod n = (c+k) \bmod n$.

11.2 The GRU Failure on $\mathbb{Z}/5$

Train a 1-layer GRU at $d = 64$ on $\mathbb{Z}/5$ (same configuration as the $d = 64, L = 1$ cell of Table 6) and run the probe. Table 7 summarizes.

Table 7: Trajectory diagnostics on $\mathbb{Z}/5$. The two architectures are trained with the same task, training depths, and number of epochs.

	GRU $d = 64$	PHASESUMNET $d = 16$
OOD accuracy ($k > 5$)	0.23	1.000
State magnitude at $k > 5$	6.75	4.00
Magnitude ratio $\ h(30)\ /\ h(5)\ $	1.19	1.00
State $\ \Delta\ /\ h\ $ (relative)	0.17	0.98
Logit $\ \Delta\ $	9.87	0.46
Argmax invariance under $5 \cdot \tau$	0.54	1.000

State-level closure is misleading. The GRU’s relative state movement under $5 \cdot \tau$ is only 0.17 (state changes by 17% of its magnitude), while PHASESUMNET’s is 0.98 (state changes by nearly its full magnitude). On state closure alone the GRU looks better. But its argmax invariance is 0.54 and PHASESUMNET’s is 1.00.

The resolution is that the readout-relevant quantity is logits, not states. PHASESUMNET’s state lives on the unit circle in $\mathbb{R}^{2d}$, and the readout’s null space (under addition of $n \cdot \theta(\tau) \equiv 0 \pmod{2\pi}$) is exactly the direction of state motion. State moves; logits don’t. The GRU’s state geometry has no such alignment: small state shifts get amplified by the linear readout into $22\times$ larger logit shifts (9.87 vs 0.46).

The per- k pattern reveals the failure mode. Table 8 shows the GRU’s argmax invariance at each k .

Invariance is high near the training boundary ($k = 0, 1$), catastrophically zero in the transition zone ($k = 3\text{--}7$), and recovers at large k . The recovery is misleading: at $k = 20\text{--}25$ the GRU’s state magnitude has saturated against the tanh ceiling ($\|h\|$ grows from 5.12 at $k = 0$ to 7.04 at $k = 30$ and asymptotes), and the logits become a near-constant function of k (logit $\|\Delta\|$ falls from 24.96 at $k = 5$ to 2.05 at $k = 20$). The prediction is constant in k , not group-respecting.

This is the unified mechanism for the $\mathbb{Z}/5$ failure: *the state saturates against the tanh ceiling past the training depth, making the readout a near-constant function of k , which is right $1/n$ of the*

Table 8: Per- k argmax invariance under $5 \cdot \tau$ for the $\mathbb{Z}/5$ GRU.

k	invariance	k	invariance
0	0.83	12–15	0.50–0.61
1	0.63	16–19	0.67–0.80
2	0.28	20–25	0.85–0.98
3–7	0.00		
8–11	0.38		

time. No amount of capacity within the tested range escapes this, because the boundedness is a property of the activation, not the layer width.

11.3 The $\mathbb{Z}/7$ Grokking Phenomenon

We applied the same trajectory diagnostics to the 2-layer GRU at $d = 128$ on $\mathbb{Z}/7$ (the partial-escape cell). One seed of this configuration was run with the same hyperparameters as Experiment 5 (120k samples, 25 epochs, $\text{lr} = 3 \times 10^{-3}$). The training trajectory is shown in Table 9.

Table 9: Training trajectory for 2-layer GRU $d = 128$ on $\mathbb{Z}/7$. Training loss reaches zero at step 2000 and stays there. OOD accuracy sits at ~ 0.70 for 4000 more steps, then jumps at step 6000.

Step	Train loss	OOD acc
500	0.0005	0.632
2000	0.0001	0.702
5500	0.0000	0.710
6000	0.0005	0.940
8000	0.0000	0.947
11700	0.0000	0.947

The pattern of (i) training loss converging to zero, (ii) OOD plateauing for an extended period, then (iii) abrupt OOD improvement, is the standard signature of grokking (Power et al., 2022). At step 6000, with the model having been at zero loss for ~ 4000 steps, OOD jumps from 0.71 to 0.94. The brief loss spike at the same step is consistent with the optimization stepping into a qualitatively different basin.

The trajectory diagnostics on the grokked model are striking (Table 10).

Table 10: Trajectory diagnostics on $\mathbb{Z}/7$ for the grokked 2-layer GRU, compared to the failing baseline ($\mathbb{Z}/5$ 1-layer GRU) and the success reference (PHASESUMNET on $\mathbb{Z}/7$).

	1-layer GRU $\mathbb{Z}/5$ (fails)	2-layer GRU $\mathbb{Z}/7$ (grokked)	PHASESUMNET $\mathbb{Z}/7$ (suc)
OOD accuracy	0.23	0.947	1.000
Magnitude ratio (30 / 5)	1.19	1.02	1.00
State $\ \Delta\ $ relative	0.13	0.17	0.98
Logit $\ \Delta\ $	9.87	3.30	0.46
Argmax invariance	0.58	0.86	1.00

The grokked GRU has stopped saturating. Magnitude ratio drops from 1.19 (failing) to 1.02 (grokked); state magnitude is nearly constant across k . The model has found an approximately-periodic state trajectory that bypasses the tanh ceiling.

The grokked GRU has partially aligned its readout. Logit $\|\Delta\|$ falls $3\times$ (from 9.87 to 3.30); argmax invariance climbs from 0.58 to 0.86. The model has moved most of the way from the failing baseline to the unit-circle success.

11.4 What Grokking Found, and What It Did Not

We interpret the grokked GRU as having discovered, via late-training optimization, an approximate version of what PHASESUMNET implements by construction: a bounded state trajectory along directions the readout partially ignores modulo n . Three pieces of evidence support this reading:

1. The magnitude ratio is essentially identical to PHASESUMNET’s (1.02 vs 1.00): the state has become effectively bounded.
2. Logit $\|\Delta\|$ has dropped by a factor of 3, indicating the readout has lost most of its amplification of state shifts.
3. The phase transition occurs at step 6000—4000 steps after training loss has converged—which is the signature of slow discovery of a structured solution, not faster convergence.

The grokked solution is not exact: argmax invariance is 0.86 vs PHASESUMNET’s 1.00, and OOD is 0.947 vs 1.000. The GRU has *approximately* discovered $\mathbb{Z}/7$ representation, not exactly. We hypothesize that with longer training or larger capacity it might approach 1.000, but at substantial parameter and training cost relative to PHASESUMNET’s 551 parameters reaching 1.000 by step 1000.

11.5 Implications for the $\mathbb{Z}/5$ Claim

Does the $\mathbb{Z}/7$ grokking finding undermine the $\mathbb{Z}/5$ structural claim? We believe not, because:

- In 48 $\mathbb{Z}/5$ runs spanning a $285\times$ capacity range, three seeds each, no run shows the late-training jump characteristic of grokking. All saturate within ~ 1500 steps and remain at the chance plateau.
- The diagnostic signature of saturation (monotonically growing $\|h\|$, constant argmax at large k) is present in every $\mathbb{Z}/5$ GRU we examined and absent in the grokked $\mathbb{Z}/7$ model.

The most defensible version of the structural claim is therefore: *within practical training budgets, real-valued gated recurrence does not learn to represent $\mathbb{Z}/n$ for $n \geq 5$ reliably. At $n = 7$ with two layers and large width, a fraction of training runs grok to an approximate representation late in training, producing the partial escape observed in Table 5. Whether $\mathbb{Z}/5$ admits grokking at sufficiently long training or larger capacity is open.*

12 Discussion

12.1 What We Can Claim

1. **Additive-linear architectures cannot represent $\mathbb{Z}/n$ for $n \geq 3$.** Provable (Theorem 3), confirmed empirically across $n = 3, 5, 7, 11, 13$ at accuracy exactly $1/n$.

2. **Softmax attention is additive composition.** The complex transformer in Experiment 1 learns the algebra ($|\Delta| = 2.43$ radians, target π) but cannot compose it. This is consistent with Hahn (2020) and the formal-language literature.
3. **The unit circle in $\mathbb{C}$ admits a faithful representation of every $\mathbb{Z}/n$ in one dimension.** PHASESUMNET reaches OOD 1.000 ± 0.000 at every n we tested, with 306–845 parameters.
4. **Bounded real-valued gated recurrence does not solve $\mathbb{Z}/n$ for $n \geq 5$ within practical training budgets.** $285\times$ capacity sweep, two architecture families, three seeds, all at ≤ 0.34 OOD. The mechanism is state saturation against the tanh ceiling.
5. **Grokking can partially leak the barrier on $\mathbb{Z}/7$.** With 2-layer GRU at $d = 128$, a late-training phase transition sometimes produces an approximate $\mathbb{Z}/7$ representation (OOD ~ 0.95). The grokked model has stopped saturating—a structurally distinct regime from the failing baseline.
6. **The qualitative separation is between architectures matching the symmetry by construction and those discovering it through optimization,** including potentially via grokking. The former generalize predictably with $O(n)$ parameters; the latter require orders of magnitude more parameters and may not always discover the structure within standard training budgets.

12.2 What We Cannot Yet Claim

1. Whether $\mathbb{Z}/5$ admits grokking with $\gg 25$ epochs of training or $\gg 1\text{M}$ parameters. The signature is absent in our 48 cells but the test is finite.
2. Whether the saturation mechanism is causally responsible for the $\mathbb{Z}/5$ failure. The clean test is to replace the tanh-bounded RNN with an unbounded normalized one; the prediction is that the saturation pattern should disappear and OOD should change. We have not run this experiment.
3. Whether the $n=7$ grokking phenomenon scales to $n \in \{11, 13\}$ at sufficient capacity. Experiment 4 measured 2-layer GRU at small d on these and saw chance; we have not retested at $d \in \{128, 256\}$.
4. Anything about real-language negation. Our experiments use controlled cyclic-rotation tasks. Real-language phenomena are governed by the cyclic structure of negation but also by scope, modality, sarcasm, and idiomatic constructions that we do not model.

12.3 Implications for Practitioners

The findings suggest a design heuristic. For any task whose underlying structure can be identified with a known group (cyclic groups for parity-like tasks; $\text{SO}(3)$ for 3D pose; $\text{SE}(3)$ for rigid body motion; permutations for set processing), an architecture whose state geometry contains a faithful representation of that group should be preferred over a generic architecture. The cost in expressiveness from constraining the state geometry is offset by orders-of-magnitude better sample efficiency, deterministic generalization properties, and the absence of late-training phase transitions.

For cyclic-group structure specifically: complex unit-circle representations are the minimal sufficient construction. Real-valued gated recurrences are insufficient for $n \geq 5$ within practical training budgets; transformers with softmax attention are insufficient at any n .

12.4 Connection to Existing Theory

Our negative result for softmax attention (Corollary 8) is a special case of Hahn (2020)’s impossibility result for parity in transformer self-attention. We extend it to general cyclic groups.

Our positive result for unit-circle architectures relates to the broader literature on group-equivariant networks (Cohen and Welling, 2016). The novelty in our setting is the *minimality* of the construction: the same algebraic structure that motivates large-scale group-equivariant CNNs for image data is sufficient to solve a sequence task with 300–800 parameters.

The grokking finding for the $\mathbb{Z}/7$ 2-layer GRU is consistent with the broader grokking literature in suggesting that overparameterized models can discover structured solutions late in training. We believe the cleanness of the cyclic-group setting may make it a useful benchmark for mechanistic studies of grokking.

13 Limitations

1. **Synthetic task family.** All experiments are on controlled cyclic-group rotation tasks. Real-world phenomena with cyclic structure (musical pitch, days of week, cardinal directions, negation in natural language) are noisier and partially observable.
2. **Limited cyclic-group orders.** We test $n \leq 13$. Whether the pattern holds for $n = 100$ or $n = 1000$ is unknown.
3. **Three seeds in the capacity sweep.** The 2-layer GRU on $\mathbb{Z}/7$ grokking happens in roughly one seed out of three at the $d = 128$ scale; a 10-seed rerun would tighten the variance.
4. **Single architecture per category.** We test one complex transformer, one GRU, one LSTM. There are many transformer variants (Su et al. (2024)’s RoPE, ALiBi, etc.) we have not tested.
5. **Limited mechanism analysis.** The trajectory diagnostics focus on state and logit dynamics in the last position. A finer analysis (per-position state evolution, gate values, etc.) would better characterize what the gated models do and do not internalize.

14 Conclusion

We have studied cyclic-group composition tasks as a controlled benchmark for architectural inductive bias. Combining a formal proof that additive-linear architectures cannot represent $\mathbb{Z}/n$ for $n \geq 3$, an empirical demonstration that softmax transformers fall into the additive class, a positive existence result for a 300-parameter unit-circle architecture that generalizes across $n \in \{2, 3, 5, 7, 11, 13\}$, and a mechanistic analysis of where the real-valued gated baselines fail, we arrive at the following synthesis:

Architectures whose composition operator is a group homomorphism into a state space containing the task’s symmetry group generalize on cyclic-group tasks predictably and with $O(n)$ parameters. Architectures whose composition operator is additive or whose state saturates do not, regardless of capacity. Between these extremes sits a regime—two-layer GRU at large width on $\mathbb{Z}/7$ —where the architecture can in principle discover the group structure but must do so through a late-training phase transition (grokking) that is unreliable and parameter-expensive. The qualitative split between *architectural match by construction* and *discovery through optimization* appears to be the central distinction.

We hope the framework—cyclic-group tasks as a graded diagnostic, trajectory diagnostics as a mechanistic probe, and the structural vs. grokked vs. constructed taxonomy—is useful for future work on architectural inductive bias.

Code and data. All code, training logs, and per-seed results are available at <https://github.com/Minds-R-Lab/ComplexAttn>.

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